Type MS · Identify parts of the eye and state their functions

Answer 1 of 14 · Question 2 part a.ii · 3 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

2	(a)	(i)	• lens in correct position; • attached both ends to suspensory ligaments;	MP1- accept lens between ciliary muscles/diameter more than width of ciliary muscle MP2- accept suspensory ligament attached to ciliary muscle	2
		(ii)	A = cornea; B = retina; C = optic nerve;		3
	(b)		• ciliary muscles relax; • suspensory ligaments tighten/taut; • they pull on lens/lens stretched; • lens is thin(er)/flat(er/less convex); • less refraction/bending of (light)	Mp2- ignore suspensory ligament contract MP5= do not accept less light refracted	4
					Total = 9

Type MS · Identify parts of the eye and state their functions

Answer 1 of 14 · Question 2 part a.ii · 3 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify parts of the eye and state their functions

Answer 2 of 14 · Question 3 part a · 1 mark

2019 JUN · Paper 01 · find part a in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Identify parts of the eye and state their functions

Answer 3 of 14 · Question 1 part a · 5 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

1(a)	line labelled A pointing to retina; line labelled B pointing to ciliary muscles; line labelled C pointing to iris; line labelled D pointing to optic nerve; line labelled E pointing to eyelid;		5

Question Number	Answer	Additional guidance	Mark
1(b)	bending at cornea; bending at lens; image formed on retina;		**3**

Question Number	Answer	Additional guidance	Mark
1(c)	long sight/myopia; short sight/hypermetropia; astigmatism;		3

Total 11 marks

Type MS · Identify parts of the eye and state their functions

Answer 3 of 14 · Question 1 part a · 5 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Identify parts of the eye and state their functions

Answer 4 of 14 · Question 3 part a.i · 1 mark

2020 JAN · Paper 02 · find part a.i in this mark scheme extract

Question number					
3	(a)	(i)	**The only correct answer is C** lens ***A** is incorrect because this is where the image is formed* ***B** is incorrect because that is what controls the size of the pupil* ***D** is incorrect because the pupil is the hole in the middle of the iris*		1
		(ii)	• retina(1) • pupil (1) • choroid(1)		3
	(b)		Value between 190 and 230 (μm)		1
	(c)	(i)	• transmit (nerve/electrical) impulses(1) • to the brain/CNS/relay neurones(1)		2
		(ii)	• less/no impulses able to reach the brain(1) • vision blurred/less clear/less focussed/image not formed/seen/reference to blindness(1)	Allow signals	2

Total for Question 3 = 9 marks

Type MS · Identify parts of the eye and state their functions

Answer 5 of 14 · Question 3 part a.ii · 3 marks

2020 JAN · Paper 02 · find part a.ii in this mark scheme extract

number				
3 (a)	(i)	**The only correct answer is C** lens ***A** is incorrect because this is where the image is formed* ***B** is incorrect because that is what controls the size of the pupil* ***D** is incorrect because the pupil is the hole in the middle of the iris*		1
	(ii)	• retina(1) • pupil (1) • choroid(1)		3
(b)		Value between 190 and 230 (μm)		1
(c)	(i)	• transmit (nerve/electrical) impulses(1) • to the brain/CNS/relay neurones(1)		2
	(ii)	• less/no impulses able to reach the brain(1) • vision blurred/less clear/less focussed/image not formed/seen/reference to blindness(1)	Allow signals	2

Total for Question 3 = 9 marks

Type MS · Identify parts of the eye and state their functions

Answer 6 of 14 · Question 6 part a · 1 mark

2021 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

number				
6	(a)	C (fovea); A/B /D have no light sensitive cells		1
	(b) (i)	Any 5 from • light becomes less/becomes darker; • reference to dim light detected by retina; • radial muscles in iris contract; • circular muscles relax; • widen/increase diameter of pupil; • to keep a constant amount of light entering the eye	Allow reference to B being in dimmer light allow more light entering the eye	5
	(ii)	• occurs when bright light is shone into eye; • too much light could damage retina; • occurs quickly to protect eye/prevent too much light entering;		3

Total 9 marks

Type MS · Identify parts of the eye and state their functions

Answer 6 of 14 · Question 6 part a · 1 mark

2021 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify parts of the eye and state their functions

Answer 7 of 14 · Question 8 part a.ii · 1 mark

2021 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

number				
8 (a)	(i)	absorbs uv light; reduces/prevents risk of skin cancer;	Allow protects against (UV) radiation	2
	(ii)	eye/ retina/ choroid/ red blood cell/ haemoglobin ;		1
(b)	(i)	mother x father Hh hh; gametes H h h; fertilisation Hh hh; phenotype white forelock normal;	Reject diagrams that show sex linkage	4
	(ii)	chance of producing a boy is 0.5/½; chance of producing offspring with condition 0.5/½; chance of producing boy with condition is ½ × ½ = ¼;	Allow 50% or equivalent anywhere in calculation Full marks for correct final answer	3
(c)		Any 4 from different (genetic) code produced; different order of amino acids; codes for different proteins/enzymes; enzyme substrate complex not formed; causes change in pigment;	Allow change in base sequence/genetic code Allow any reference to change in amino acids	4

Total 14 marks

Type MS · Identify parts of the eye and state their functions

Answer 7 of 14 · Question 8 part a.ii · 1 mark

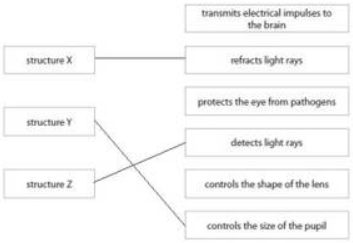
2021 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

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Type MS · Identify parts of the eye and state their functions

Answer 8 of 14 · Question 1 part a.i · 3 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

number					
1	(a)	(i)			3
		(ii)	A; B, C and D do not describe stereoscopic vision.		1
	(b)	(i)	build-up of protein/cloudy/opaque; in the lens;		2
		(ii)	surgery/operation to replace the lens/lens replacement;		1
		(iii)	39 000^{0}000 x 43 ÷ 100; 16.77 million/16.8 million/16 770 000;		2
		(iv)	vitamin A;		1

Total 10 marks

Type MS · Identify parts of the eye and state their functions

Answer 8 of 14 · Question 1 part a.i · 3 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify parts of the eye and state their functions

Answer 9 of 14 · Question 6 part b.ii · 1 mark

2023 JAN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number					
6	(a)	(i)	Three from: • the use of embryonic stem cells is not needed/embryos not destroyed; • easily available source of cells/easy to obtain/no need to find a donor; • same cell/tissue/blood type of the patient/same antigens; • reduced chance of rejection/not recognised as foreign;		3
		(ii)	Two from: • unlicensed clinics could be used; • stem cells could develop into tumours/cancer; • risk of rejection (if the cells are not the recipients); • increased risk of infection;		2
	(b)	(i)	860 000 and 480 000; 380 000;	Ecf allow one mark for incorrectly derived figures from graph but correct answer from these figures	2
		(ii)	**D** structure Z;		1
	(c)		Three from: • cut out desired/beta-carotene/gene/gene for vitamin A; • using/cut with restriction enzymes; • insert into plasmid; • insert into/infect rice genome; • to produce golden rice;		3

Total 11 marks

Type MS · Identify parts of the eye and state their functions

Answer 9 of 14 · Question 6 part b.ii · 1 mark

2023 JAN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify parts of the eye and state their functions

Answer 10 of 14 · Question 3 part a.i · 2 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

3 (a) (i)	Part W - pupil; Part X – retina;	1 mark for each correctly named part	2
(ii)	• sends (electrical) impulses from photoreceptors/rods/cones; • to the brain/CNS; OR • sends (electrical) impulses from brain/CNS; • to iris/ciliary muscles;		2
(iii)	lens becomes thinner/flatter/less convex; less refraction;	allow less light refracted	2
(iv)	Any two from: • shape of cornea restored; • light refracted, onto the retina/correctly/notmally; • correct information sent to the brain; • image/vision not blurred/clear;		2
(b)	• measure the size/diameter of the pupil; • in a dark room; • shine a light into the eye; • measure/compare size/diameter of the pupil;	accept ref to different light intensities	4

Total Marks 12

Type MS · Identify parts of the eye and state their functions

Answer 10 of 14 · Question 3 part a.i · 2 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify parts of the eye and state their functions

Answer 11 of 14 · Question 6 part a.i · 4 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

6	(a)	(i)	• A=rods/cones/retina; • detects light; • converts light energy into electrical energy; • (changed into (nerve) impulses; • B = transmits (nerve) impulses/electrical signals; • from (retina) to brain;	At least 1 mark from A and B to get full marks	4
		(ii)	• bent inwards at cornea/lens; • rays pass through the cornea and converge after the lens; • rays cross in eye (after lens); • inverted image/arrow;	Do not accept bending outwards for MP1 and MP2 Mp3- do not accept rays crossing before the retina	4
	(b)		• use both eyes and one eye; • use of eye patch/eq; • ball thrown same distance/stated distance;; • ball thrown many times; • count how many times ball is caught; • same type/size of ball;		4
	(c)		• misshapen/eq. cornea/lens; • light not focussed correctly/ light scatters/light not refracted properly; • blurred image;		2
					Total =14

Type MS · Identify parts of the eye and state their functions

Answer 11 of 14 · Question 6 part a.i · 4 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Identify parts of the eye and state their functions

Answer 12 of 14 · Question 1 part b · 2 marks

2023 JUN · Paper 01 (regional R) · find part b in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Identify parts of the eye and state their functions

Answer 13 of 14 · Question 1 part a · 6 marks

2023 JUN · Paper 01 (regional R) · find part a in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Identify parts of the eye and state their functions

Answer 14 of 14 · Question 1 part a · 7 marks

2023 NOV · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

1 (a)	<table><tr><th>Description</th><th>Part</th></tr><tr><td>contains light sensitive cells</td><td>retina;</td></tr><tr><td>vibrates when receives sound waves</td><td>ear drum;</td></tr><tr><td>helps the body balance</td><td>semi-circular canals;</td></tr><tr><td>controls the amount of light entering the eye</td><td>pupil;</td></tr><tr><td>transfers vibrations across the middle ear</td><td>ossicles;</td></tr><tr><td>turns vibrations into nerve impulses</td><td>cochlea;</td></tr><tr><td>refracts light entering the eye</td><td>cornea;</td></tr></table>		7
(b)	• stereoscopic/3D vision; • each eye gives a slightly different image/view; • allows better distance/speed judgement/wider field of view;	Allow better judgement of depth for 3D vision	3

Total 7 marks

Type MS · Identify parts of the eye and state their functions

Answer 14 of 14 · Question 1 part a · 7 marks

2023 NOV · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks